Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	LTVIP2026TMIDS82222
Project Name	HouseHunt – Finding Your Perfect Rental Home
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form (Name, Email, Password, Role – Renter/Owner)
		Password Encryption using bcrypt
		Store user data in MongoDB
FR-2	User Authentication	Login using Email & Password
		JWT Token Generation
		Role-based Access (Admin / Owner / Renter)
FR-3	Owner Approval System	Owner requests admin approval
		Admin verifies and approves/rejects owner
FR-4	Property Management (Owner)	Add Property (Type, Address, Rent, Images)
		Edit Property Details
		Delete Property
		Update Availability Status
FR-5	Property Browsing (Renter)	View All Available Properties
		Apply Filters (Location, Rent Range, Type)
		View Property Details
FR-6	Booking Management	Send Booking Request
		Owner Approves/Rejects Booking
		Booking Status Tracking (Pending/Approved/Rejected)
FR-7	Admin Management	View All Users
		Approve Owner Accounts
		Monitor Property Listings
FR-8	Dashboard System	Renter Dashboard (Bookings & Status)
		Owner Dashboard (Properties & Requests)
		Admin Dashboard (User Control Panel)

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system must provide a simple, user-friendly interface for renters, owners, and admin with easy navigation and clear dashboards.
NFR-2	Security	Password encryption using bcrypt, JWT-based authentication, role-based authorization, and secure API access.
NFR-3	Reliability	The system should handle booking requests accurately without data loss and maintain consistent database transactions.
NFR-4	Performance	The application should load property listings quickly and respond to API requests within minimal time.
NFR-5	Availability	The system should be available 24/7 for users to browse properties and manage bookings.
NFR-6	Scalability	The application should support increasing users and property listings without performance degradation (MongoDB scalability support).
NFR-7	Maintainability	Code structure should follow MVC architecture for easy updates and debugging.
NFR-8	Compatibility	The system should work across modern web browsers and devices (Desktop & Mobile responsive UI).